

B6.3 Monitoring and maintaining health

Summary

Diseases affect the health of populations of both humans and plants. Scientists are constantly on the lookout for ways of preventing and combating disease. The prevention of disease in plants is important so that we are able to grow healthy plants enabling us to feed ourselves and enhance our environment. The understanding of how disease is spread, how our bodies defend themselves against disease and how immunity is achieved is essential to enable us to combat potentially fatal diseases spreading throughout whole populations. Non-communicable diseases also have an impact on the health of the population. The prevention of these diseases is frequently discussed in the media, with advice being given to us on how to reduce our risk of contracting these diseases through our life-style choices and discussion of new technologies.

Underlying knowledge and understanding

Learners should be familiar with the effects of 'recreational' drugs (including substance misuse) on behaviour, health and life processes, the impact of exercise, asthma and smoking on the gas exchange system and the consequences of imbalances in the diet, including obesity, starvation and deficiency diseases.

Common misconceptions

Research has shown that learners tend to view all micro-organisms as being non-beneficial. They tend to consider health as just physical and do not consider mental health. Learners also confuse which diseases are inherited and which are caught. They see cancer as a genetic disease.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers. All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
BM6.3i	translate information between graphical and numerical forms	M4a
BM6.3ii	construct and interpret frequency tables and diagrams, bar charts and histograms	M2c
BM6.3iii	understand the principles of sampling as applied to scientific data	M2d
BM6.3iv	use a scatter diagram to identify a correlation between two variables	M2g
BM6.3v	calculate cross-sectional areas of bacterial cultures and clear agar jelly using πr^2	M5c

Topic content		Opportunities to cover:		Practical suggestions
Learning outcomes	To include	Maths	Working scientifically	
B6.3a describe the relationship between health and disease				
B6.3b describe different types of diseases	communicable and non-communicable diseases			
B6.3c describe the interactions between different types of disease	HIV and tuberculosis; HPV and cervical cancer	M4a		
B6.3d explain how communicable diseases (caused by viruses, bacteria, protists and fungi) are spread in animals and plants	scientific quantities, number of pathogens, number of infected cases, estimating number of cases	M2c, M2g	WS1.4b	
B6.3e explain how the spread of communicable diseases may be reduced or prevented in animals and plants	detection of the antigen, DNA testing, visual identification of the disease	M2c	WS1.4b	
B6.3f describe a minimum of one common human infection, one plant disease and sexually transmitted infections in humans including HIV/AIDS	human infections: one example of each viral, fungal, bacterial plant diseases: viral tobacco mosaic virus TMV, fungal <i>Erysiphe graminis</i> barley powdery mildew, bacterial <i>Agrobacterium tumefaciens</i> crown gall disease			
B6.3g ☑ describe physical plant defence responses to disease	leaf cuticle, cell wall			
B6.3h ☑ describe chemical plant defence responses	antimicrobial substances			
B6.3i ☑ describe different ways plant diseases can be detected and identified, in the lab and in the field	the laboratory detection of the DNA or antigen from the disease causing organism. The field diagnosis by observation and microscopy			
B6.3j explain how white blood cells and platelets are adapted to their defence functions in the blood				

B6.3k	describe the non-specific defence systems of the human body against pathogens				
B6.3l	explain the role of the immune system of the human body in defence against disease				
B6.3m ☒	describe how monoclonal antibodies are produced			WS1.1d	
B6.3n ☒	describe some of the ways in which monoclonal antibodies can be used	their role in detecting antigens in pregnancy testing, detection of diseases (prostate cancer) and potentially treating disease (targeting cancer cells)			
B6.3o	explain the use of vaccines and medicines in the prevention and treatment of disease	antibiotics, antivirals and antiseptics		WS1.1g, WS1.1h	Research into whether children should be routinely vaccinated?
B6.3p ☒	explain the aseptic techniques used in culturing organisms	use of alcohol, flaming, autoclaving of glassware and growth media, and measures used to stop contaminants falling onto/into the growth media (e.g. working around a Bunsen burner)	M3d, M5c	WS1.1h, WS1.2c, WS2a, WS2b, WS2c, WS2d	Investigation into growth bacterial cultures using aseptic techniques. (PAG B1, PAG B7)
B6.3q	describe the processes of discovery and development of potential new medicines	preclinical and clinical testing	M2d, M3d, M5c	WS1.1d, WS2a, WS2b, WS2c, WS2d	Investigation into growth bacterial cultures using aseptic techniques. (PAG B1, PAG B7)
B6.3r	recall that many non-communicable human diseases are caused by the interaction of a number of factors	cardiovascular diseases, many forms of cancer, some lung (bronchitis) and liver (cirrhosis) diseases and diseases influenced by nutrition, including type 2 diabetes			
B6.3s	evaluate some different treatments for cardiovascular disease	lifestyle, medical and surgical	M2g, M2d		

B6.3t	analyse the effect of lifestyle factors on the incidence of non-communicable diseases at local, national and global levels	lifestyle factors to include exercise, diet, alcohol and smoking	M2c, M2d, M4a		
B6.3u	describe cancer as the result of changes in cells that lead to uncontrolled growth and division				
B6.3v	discuss potential benefits and risks associated with the use of stem cells in medicine	tissue transplantation and rejection		WS1.1c, WS1.1d, WS1.1e, WS1.1f, WS1.1g, WS1.1h, WS1.1j	
B6.3w	explain some of the possible benefits and risks of using gene technology in medicine	practical and ethical considerations		WS1.1c, WS1.1d, WS1.1e, WS1.1j	
B6.3x	discuss the potential importance for medicine of our increasing understanding of the human genome	the ideas of predicting the likelihood of diseases occurring and their treatment by drugs which are targeted to genomes		WS1.1c, WS1.1d, WS1.1j	